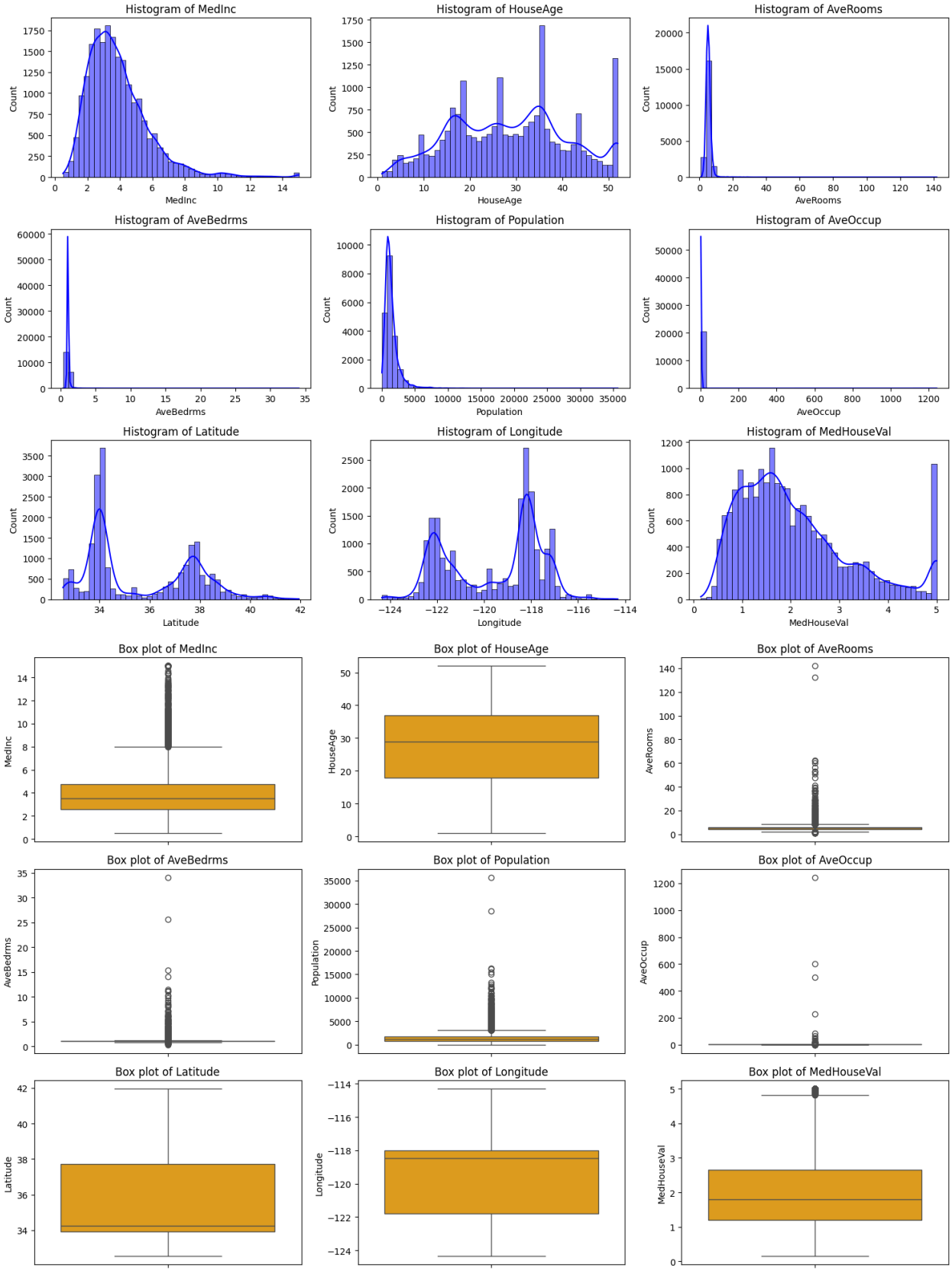
Open in Colab

In [ ]:
```
#lab 1
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.datasets import fetch_california_housing
data = fetch_california_housing(as_frame=True)
housing_df=data.frame
numerical_features=housing_df.select_dtypes(include=[np.number])
numerical_features.head()

plt.figure(figsize=(15,10))
for i,feature in enumerate(numerical_features.columns):
    plt.subplot(3,3,i+1)
    sns.histplot(data=housing_df[feature],kde=True,bins=45,color='blue')
    plt.title(f'Histogram of {feature}')
plt.tight_layout()
plt.show()

plt.figure(figsize=(15,10))
for i,feature in enumerate(numerical_features.columns):
    plt.subplot(3,3,i+1)
    sns.boxplot(data=housing_df[feature],color='orange')
    plt.title(f'Box plot of {feature}')
plt.tight_layout()
plt.show()

print("Description of outliers")
outliers_summary={}
for feature in numerical_features.columns:
    Q1=housing_df[feature].quantile(0.25)
    Q3=housing_df[feature].quantile(0.75)
    IQR = Q3 - Q1
    lower_bound=Q1-1.5*IQR
    upper_bound=Q3+1.5*IQR
    outliers=housing_df[(housing_df[feature]<lower_bound) | (housing_df[featur
    outliers_summary[feature] = len(outliers)
print(outliers_summary)
print(housing_df.describe())
```

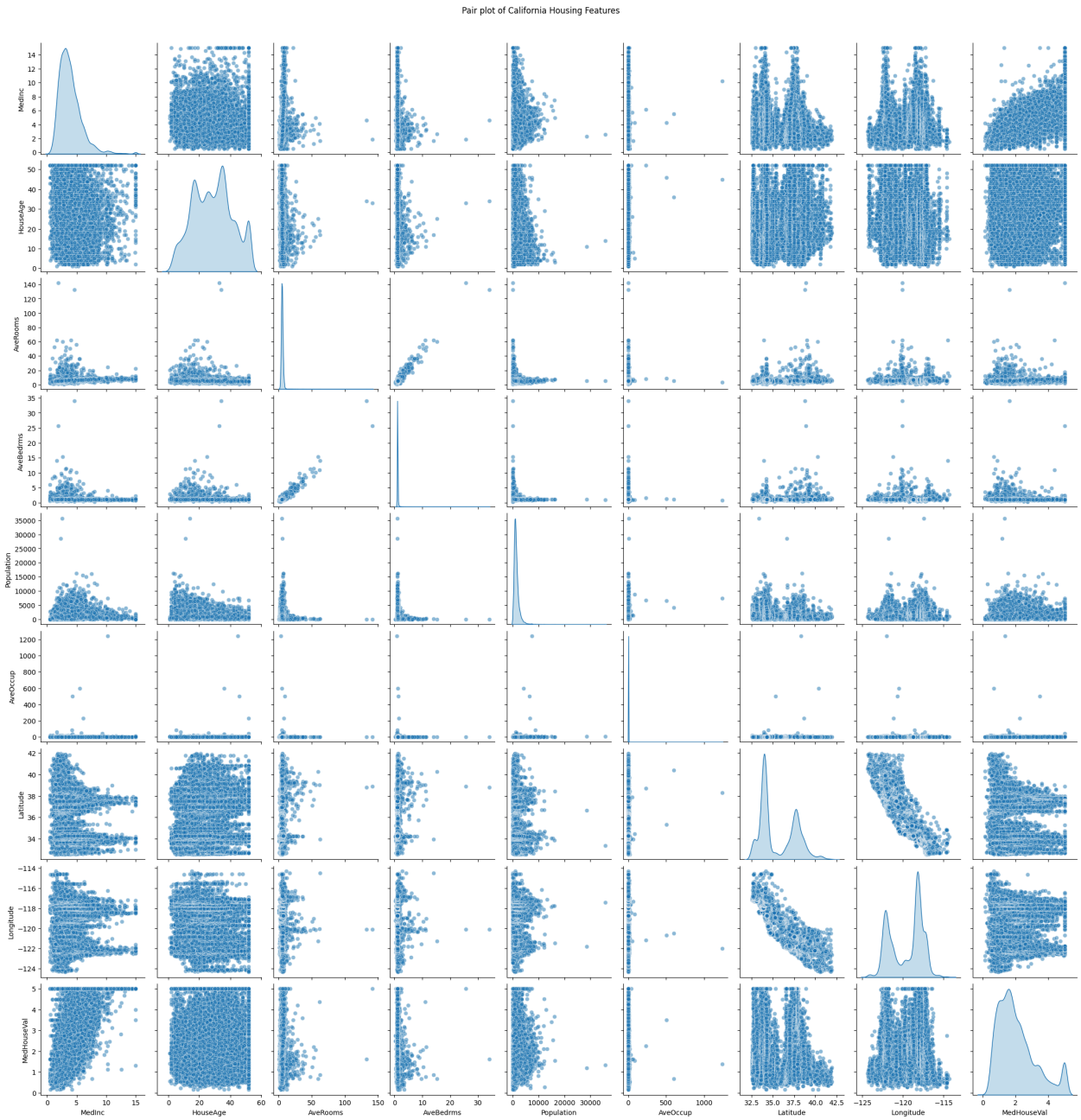

Histogram of MedInc
Count
MedInc
Histogram of HouseAge
Count
HouseAge
Histogram of AveRooms
Count
AveRooms
Histogram of AveBedrms
Count
AveBedrms
Histogram of Population
Count
Population
Histogram of AveOccup
Count
AveOccup
Histogram of Latitude
Count
Latitude
Histogram of Longitude
Count
Longitude
Histogram of MedHouseVal
Count
MedHouseVal
Box plot of MedInc
MedInc
Box plot of HouseAge
HouseAge
Box plot of AveRooms
AveRooms
Box plot of AveBedrms
AveBedrms
Box plot of Population
Population
Box plot of AveOccup
AveOccup
Box plot of Latitude
Latitude
Box plot of Longitude
Longitude
Box plot of MedHouseVal
MedHouseVal

```
Description of outliers
{'MedInc': 681, 'HouseAge': 0, 'AveRooms': 511, 'AveBedrms': 1424, 'Populatio
n': 1196, 'AveOccup': 711, 'Latitude': 0, 'Longitude': 0, 'MedHouseVal': 1071}
             MedInc      HouseAge      AveRooms     AveBedrms    Population  \
count  20640.000000  20640.000000  20640.000000  20640.000000  20640.000000
mean       3.870671     28.639486      5.429000      1.096675   1425.476744
std        1.899822     12.585558      2.474173      0.473911   1132.462122
min        0.499900      1.000000      0.846154      0.333333      3.000000
25%        2.563400     18.000000      4.440716      1.006079    787.000000
50%        3.534800     29.000000      5.229129      1.048780   1166.000000
75%        4.743250     37.000000      6.052381      1.099526   1725.000000
max       15.000100     52.000000    141.909091     34.066667  35682.000000

           AveOccup      Latitude     Longitude   MedHouseVal
count  20640.000000  20640.000000  20640.000000  20640.000000
mean       3.070655     35.631861   -119.569704      2.068558
std       10.386050      2.135952      2.003532      1.153956
min        0.692308     32.540000   -124.350000      0.149990
25%        2.429741     33.930000   -121.800000      1.196000
50%        2.818116     34.260000   -118.490000      1.797000
75%        3.282261     37.710000   -118.010000      2.647250
max     1243.333333     41.950000   -114.310000      5.000010
```

In [ ]:

```
#lab 2
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.datasets import fetch_california_housing
california_data=fetch_california_housing(as_frame=True)
data=california_data.frame
correlation_matrix=data.corr()
plt.figure(figsize=(10,8))
sns.heatmap(correlation_matrix,annot=True,cmap='BuPu',fmt='.2f',linewidths=1.5
plt.title('Correlation Matrix of California housing features')
plt.show()
sns.pairplot(data,diag_kind='kde',plot_kws={'alpha':0.5})
plt.suptitle('Pair plot of California Housing Features',y=1.02)
plt.show()
```

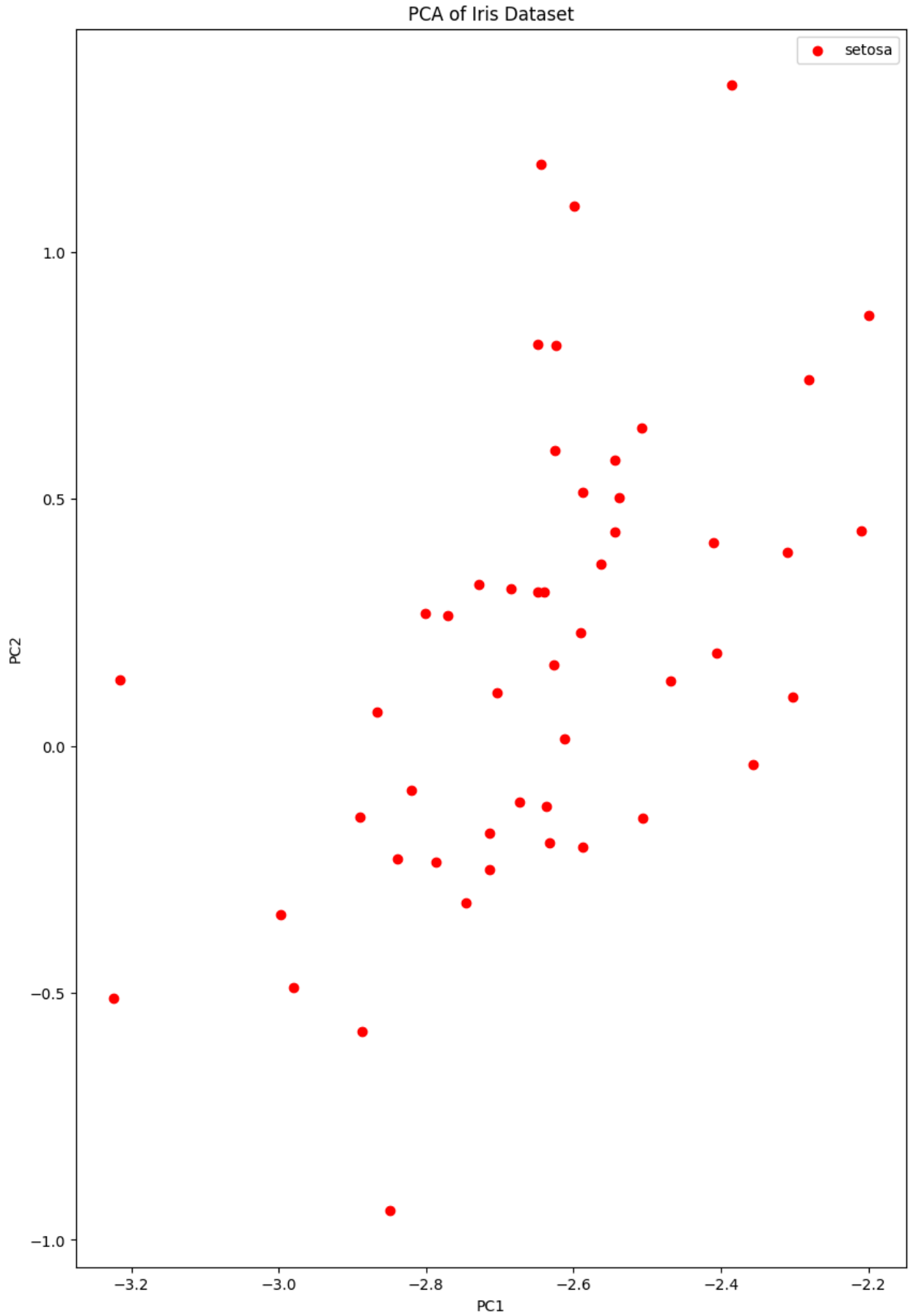
Correlation Matrix of California housing features
MedInc
HouseAge
AveRooms
AveBedrms
Population
AveOccup
Latitude
Longitude
MedHouseVal
1.00 -0.12 0.33 -0.06 0.00 0.02 -0.08 -0.02 0.69
-0.12 1.00 -0.15 -0.08 -0.30 0.01 0.01 -0.11 0.11
0.33 -0.15 1.00 0.85 -0.07 -0.00 0.11 -0.03 0.15
-0.06 -0.08 0.85 1.00 -0.07 -0.01 0.07 0.01 -0.05
0.00 -0.30 -0.07 -0.07 1.00 0.07 -0.11 0.10 -0.02
0.02 0.01 -0.00 -0.01 0.07 1.00 0.00 0.00 -0.02
-0.08 0.01 0.11 0.07 -0.11 0.00 1.00 -0.92 -0.14
-0.02 -0.11 -0.03 0.01 0.10 0.00 -0.92 1.00 -0.05
0.69 0.11 0.15 -0.05 -0.02 -0.02 -0.14 -0.05 1.00
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75

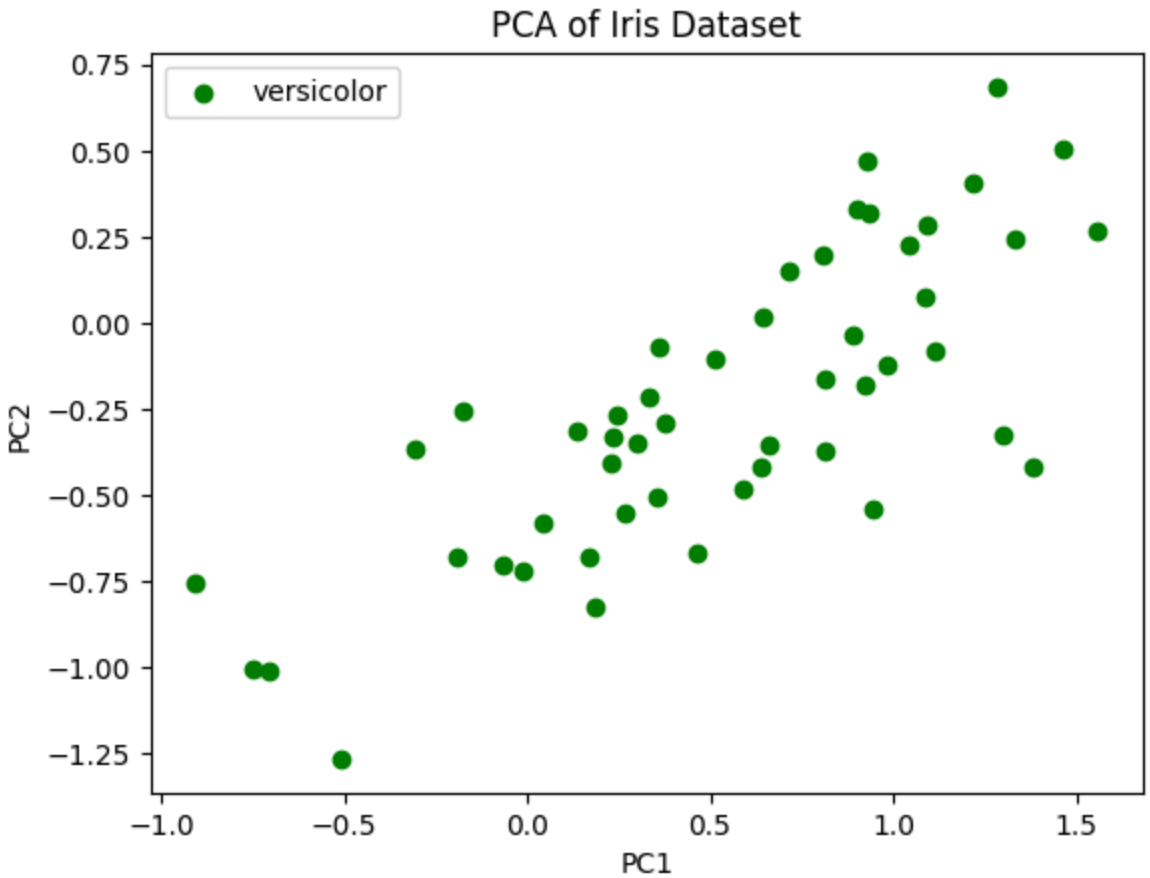


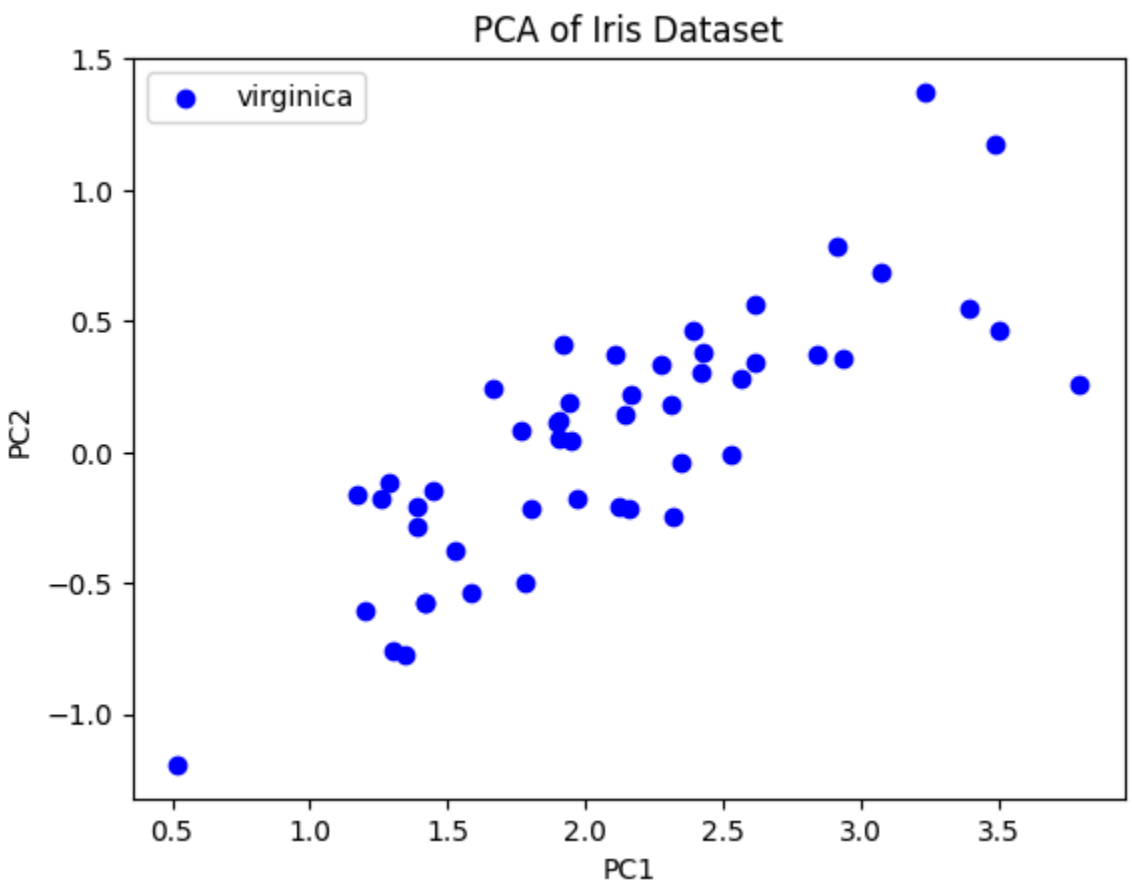
In [ ]:

```
#lab 3
import pandas as pd
import numpy as np
from sklearn.datasets import load_iris
from sklearn.decomposition import PCA
import matplotlib.pyplot as plt
iris=load_iris()
data=iris.data
target=iris.target
label_names=iris.target_names
iris_df=pd.DataFrame(data,columns=iris.feature_names)
pca=PCA(n_components=2)
data_reduced=pca.fit_transform(data)
reduced_df=pd.DataFrame(data_reduced,columns=['PC1','PC2'])
```

```
reduced_df['target']=target
plt.figure(figsize=(10,15))
colors=['r','g','b']
for i,target in enumerate(np.unique(target)):
    plt.scatter(reduced_df[reduced_df['target']==target]['PC1'],reduced_df[rec
    plt.xlabel('PC1')
    plt.ylabel('PC2')
    plt.title('PCA of Iris Dataset')
    plt.legend()
    plt.show()
```


PCA of Iris Dataset
setosa
1.0
0.5
0.0
−0.5
−1.0
PC2
−3.2
−3.0
−2.8
−2.6
−2.4
−2.2
PC1


PCA of Iris Dataset
versicolor
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
−1.25
PC2
−1.0
−0.5
0.0
0.5
1.0
1.5
PC1


PCA of Iris Dataset
virginica
1.5
1.0
0.5
0.0
−0.5
−1.0
PC2
0.5
1.0
1.5
2.0
2.5
3.0
3.5
PC1

In [ ]:
```python
#lab 4
from google.colab import files
uploaded = files.upload()

import pandas as pd
import numpy as np
data=pd.read_csv('/content/enjoysport.csv')
attribute=np.array(data)[:,:-1]
target=np.array(data)[:,-1]
print("Training Data:\n")
print(data)

def train(att,tar):
  for i,val in enumerate(tar):
    if val=='yes':
      specific_h=att[i].copy()
      break
  for i,val in enumerate(att):
    if tar[i]=='yes':
      for x in range(len(specific_h)):
        if val[x]!=specific_h[x]:
          specific_h[x]='?'
        else:
          pass
  return specific_h

print("\n")
print("Hypothesis from training:\n")
print(train(attribute,target))
```

Browse...

Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

```
Saving enjoysport.csv to enjoysport (1).csv
Training Data:

     sky air_temp humidity    wind water forecast enjoy_sport
0  sunny     warm   normal  strong  warm     same         yes
1  sunny     warm     high  strong  warm     same         yes
2  rainy     cold     high  strong  warm   change          no
3  sunny     warm     high  strong  coid   change         yes


Hypothesis from training:

['sunny' 'warm' '?' 'strong' '?' '?']
```

In [ ]:
```python
#lab 5
import numpy as np
from sklearn.neighbors import KNeighborsClassifier
data=np.random.rand(100).reshape(-1,1)
labels=np.array(["Class1" if x<=0.5 else "Class2" for x in data [:50]])

train_data,train_labels=data[:50],labels
```

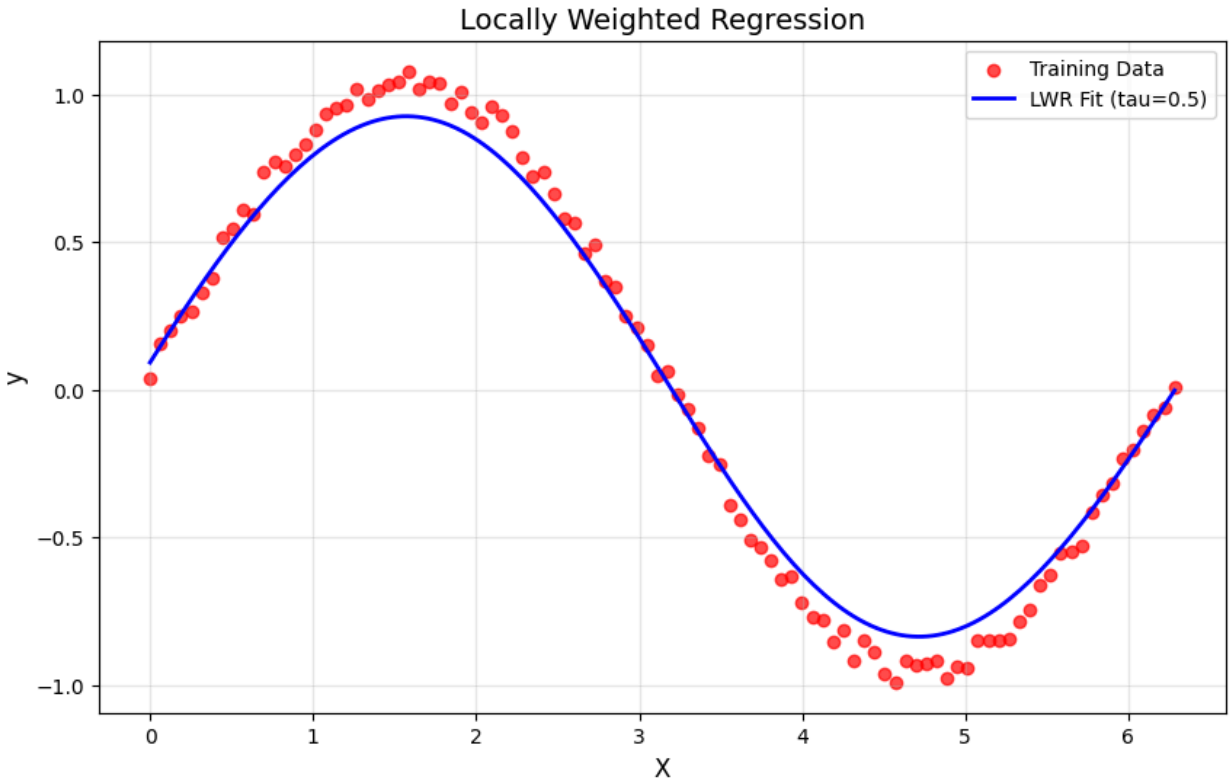
```python
test_data=data[50:]

k_values=[1,2,3,4,5,20,30]
for k in k_values:
  knn=KNeighborsClassifier(n_neighbors=k)
  knn.fit(train_data,train_labels)
  predictions=knn.predict(test_data)

  print(f"Results for k={k}:")
  print(predictions,"\n")
```

```
Results for k=1:
['Class2' 'Class1' 'Class1' 'Class1' 'Class2' 'Class1' 'Class1' 'Class2'
 'Class1' 'Class1' 'Class2' 'Class1' 'Class2' 'Class1' 'Class2' 'Class2'
 'Class1' 'Class2' 'Class1' 'Class2' 'Class1' 'Class2' 'Class2' 'Class1'
 'Class1' 'Class1' 'Class2' 'Class2' 'Class1' 'Class1' 'Class2' 'Class1'
 'Class2' 'Class2' 'Class2' 'Class1' 'Class2' 'Class1' 'Class1' 'Class1'
 'Class2' 'Class1' 'Class2' 'Class1' 'Class1' 'Class1' 'Class2' 'Class2'
 'Class2' 'Class1']

Results for k=2:
['Class2' 'Class1' 'Class1' 'Class1' 'Class2' 'Class1' 'Class1' 'Class2'
 'Class1' 'Class1' 'Class2' 'Class1' 'Class2' 'Class1' 'Class2' 'Class2'
 'Class1' 'Class2' 'Class1' 'Class2' 'Class1' 'Class2' 'Class2' 'Class1'
 'Class1' 'Class1' 'Class2' 'Class2' 'Class1' 'Class1' 'Class2' 'Class1'
 'Class2' 'Class2' 'Class2' 'Class1' 'Class2' 'Class1' 'Class1' 'Class1'
 'Class2' 'Class1' 'Class2' 'Class1' 'Class1' 'Class1' 'Class2' 'Class2'
 'Class2' 'Class1']

Results for k=3:
['Class2' 'Class1' 'Class1' 'Class1' 'Class2' 'Class1' 'Class1' 'Class2'
 'Class1' 'Class1' 'Class2' 'Class1' 'Class2' 'Class1' 'Class2' 'Class2'
 'Class1' 'Class2' 'Class1' 'Class2' 'Class1' 'Class2' 'Class2' 'Class1'
 'Class1' 'Class1' 'Class2' 'Class2' 'Class1' 'Class1' 'Class2' 'Class1'
 'Class2' 'Class2' 'Class2' 'Class1' 'Class2' 'Class1' 'Class1' 'Class1'
 'Class2' 'Class1' 'Class2' 'Class1' 'Class1' 'Class1' 'Class2' 'Class2'
 'Class2' 'Class1']

Results for k=4:
['Class2' 'Class1' 'Class1' 'Class1' 'Class2' 'Class1' 'Class1' 'Class2'
 'Class1' 'Class1' 'Class2' 'Class1' 'Class2' 'Class1' 'Class2' 'Class2'
 'Class1' 'Class2' 'Class1' 'Class2' 'Class1' 'Class2' 'Class2' 'Class1'
 'Class1' 'Class1' 'Class2' 'Class2' 'Class1' 'Class1' 'Class2' 'Class1'
 'Class2' 'Class2' 'Class2' 'Class1' 'Class2' 'Class1' 'Class1' 'Class1'
 'Class2' 'Class1' 'Class2' 'Class1' 'Class1' 'Class1' 'Class2' 'Class2'
 'Class2' 'Class1']

Results for k=5:
['Class2' 'Class1' 'Class1' 'Class1' 'Class2' 'Class1' 'Class1' 'Class2'
 'Class1' 'Class1' 'Class2' 'Class1' 'Class2' 'Class1' 'Class2' 'Class2'
 'Class1' 'Class2' 'Class1' 'Class2' 'Class1' 'Class2' 'Class2' 'Class1'
 'Class1' 'Class1' 'Class2' 'Class2' 'Class1' 'Class1' 'Class2' 'Class1'
 'Class2' 'Class2' 'Class2' 'Class1' 'Class2' 'Class1' 'Class1' 'Class1'
 'Class2' 'Class1' 'Class2' 'Class1' 'Class1' 'Class1' 'Class2' 'Class2'
 'Class2' 'Class1']

Results for k=20:
['Class2' 'Class1' 'Class1' 'Class1' 'Class2' 'Class1' 'Class1' 'Class2'
 'Class1' 'Class1' 'Class2' 'Class1' 'Class2' 'Class1' 'Class2' 'Class2'
 'Class1' 'Class2' 'Class1' 'Class2' 'Class1' 'Class2' 'Class2' 'Class1'
 'Class1' 'Class1' 'Class1' 'Class1' 'Class1' 'Class1' 'Class2' 'Class1'
 'Class2' 'Class2' 'Class2' 'Class1' 'Class2' 'Class1' 'Class1' 'Class1'
 'Class2' 'Class1' 'Class1' 'Class1' 'Class1' 'Class1' 'Class2' 'Class2'
 'Class2' 'Class1']
```

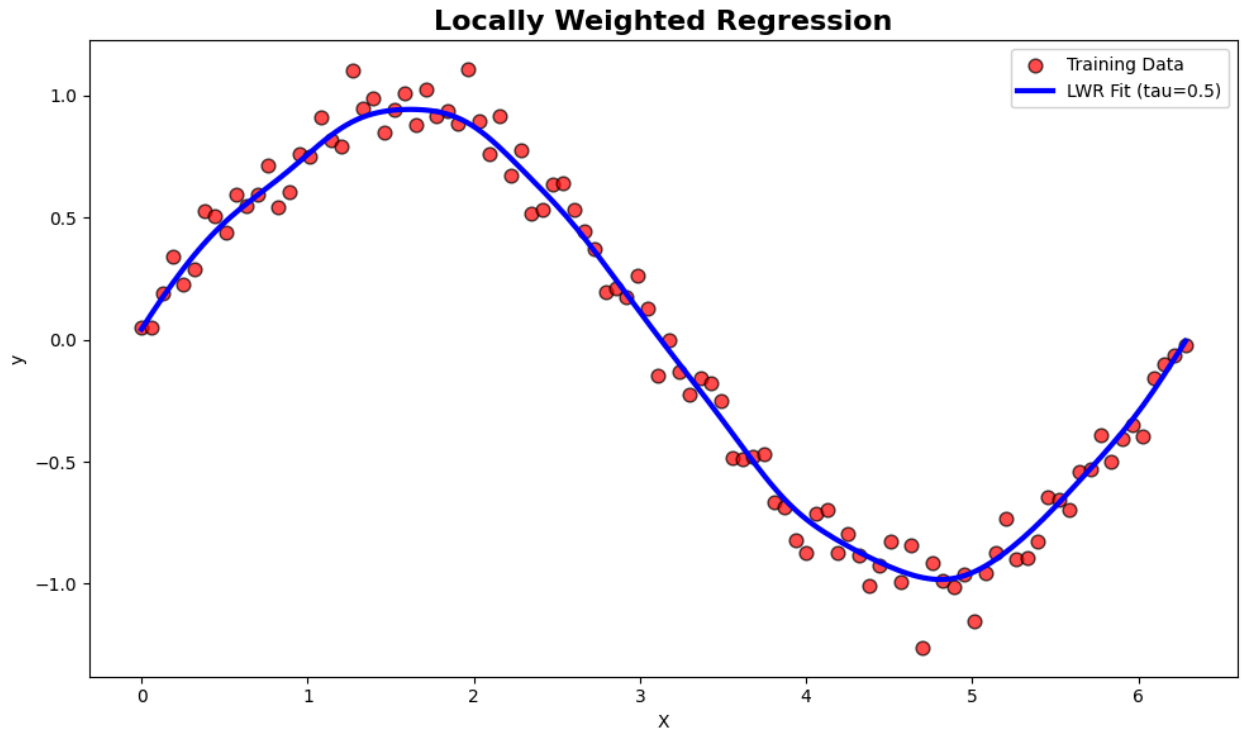
```
Results for k=30:
['Class2' 'Class1' 'Class1' 'Class1' 'Class2' 'Class1' 'Class1' 'Class2'
 'Class1' 'Class1' 'Class2' 'Class1' 'Class2' 'Class1' 'Class2' 'Class2'
 'Class1' 'Class2' 'Class1' 'Class2' 'Class1' 'Class2' 'Class2' 'Class1'
 'Class1' 'Class1' 'Class1' 'Class1' 'Class1' 'Class1' 'Class2' 'Class1'
 'Class2' 'Class2' 'Class2' 'Class1' 'Class2' 'Class1' 'Class1' 'Class1'
 'Class2' 'Class1' 'Class1' 'Class1' 'Class1' 'Class1' 'Class2' 'Class2'
 'Class2' 'Class1']
```

In [ ]:

```python
#LAB_6
import numpy as np
import matplotlib.pyplot as plt
def gaussian_kernel(x, xi, tau):
    return np.exp(-np.sum((x - xi) ** 2) / (2 * tau ** 2))
def locally_weighted_regression(x, X, y, tau):
    m = X.shape[0]
    weights = np.array([gaussian_kernel(x, X[i], tau) for i in range(m)])
    W = np.diag(weights)
    X_transpose_W = X.T @ W
    theta = np.linalg.inv(X_transpose_W @ X) @ X_transpose_W @ y
    return x @ theta
np.random.seed(42)
X = np.linspace(0, 2 * np.pi, 100)
y = np.sin(X) + 0.1 * np.random.random(100)
X_bias = np.c_[np.ones(X.shape), X]
X_test = np.linspace(0, 2 * np.pi, 200)
X_test_bias = np.c_[np.ones(X_test.shape), X_test]
tau = 0.5
y_pred = np.array([locally_weighted_regression(xi, X_bias, y, tau) for xi in X
plt.figure(figsize=(10, 6))
plt.scatter(X, y, color='red', label='Training Data', alpha=0.7)
plt.plot(X_test, y_pred, color='blue', label=f'LWR Fit (tau={tau})', linewidth
plt.xlabel('X', fontsize=12)
plt.ylabel('y', fontsize=12)
plt.title('Locally Weighted Regression', fontsize=14)
plt.legend(fontsize=10)
plt.grid(alpha=0.3)
plt.show()
```



In [ ]:
```
#lab_6_2
import numpy as np, matplotlib.pyplot as plt
from statsmodels.nonparametric.kernel_regression import KernelReg
np.random.seed(42)
X=np.linspace(0,2*np.pi,100)
y=np.sin(X)+0.1*np.random.randn(100)
model=KernelReg(y,X,'c')
x_test=np.linspace(0,2*np.pi,200); y_pred,_=model.fit(x_test)
plt.figure(figsize=(10,6))
plt.scatter(X,y,color='red',edgecolor='black',s=60,alpha=0.7,label="Training D
plt.plot(x_test,y_pred,color='blue',linewidth=3,label="LWR Fit (tau=0.5)")
plt.title("Locally Weighted Regression",fontsize=16,fontweight='bold')
plt.xlabel("X"); plt.ylabel("y")
plt.legend(frameon=True)
plt.tight_layout()
plt.show()
```

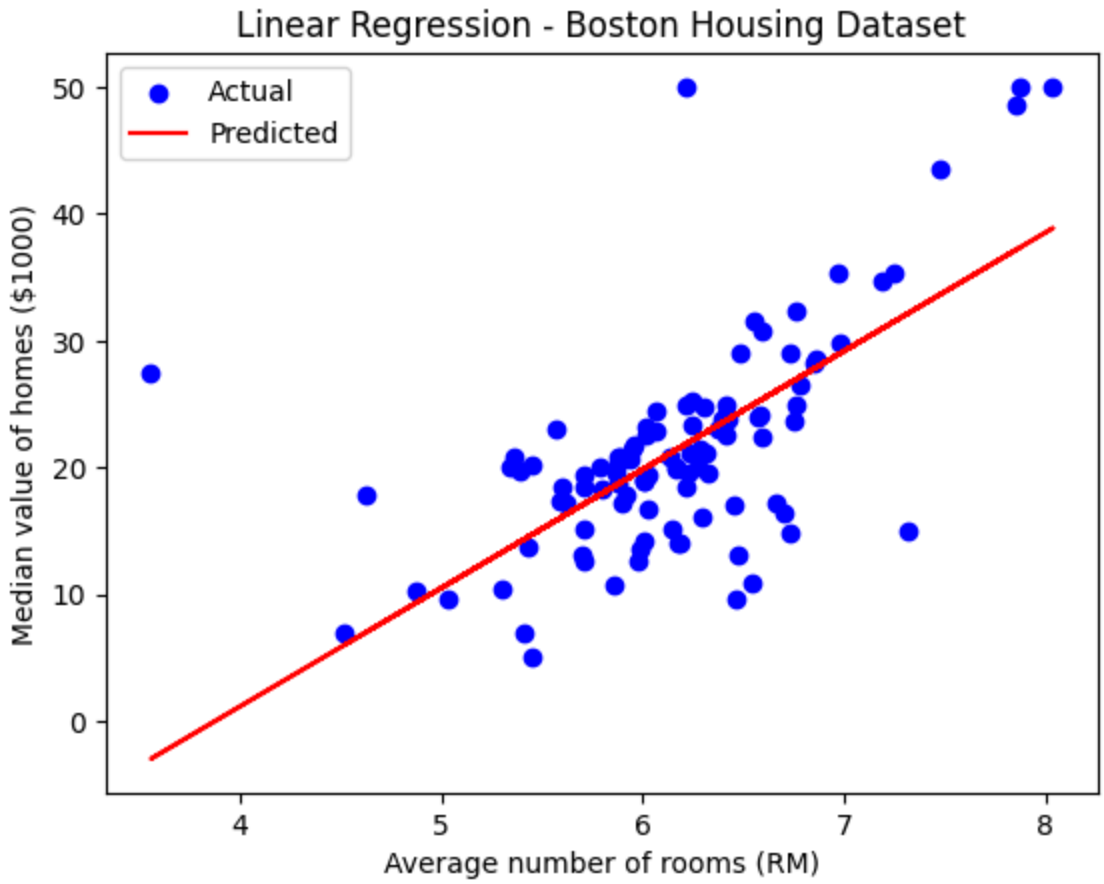


In [ ]:

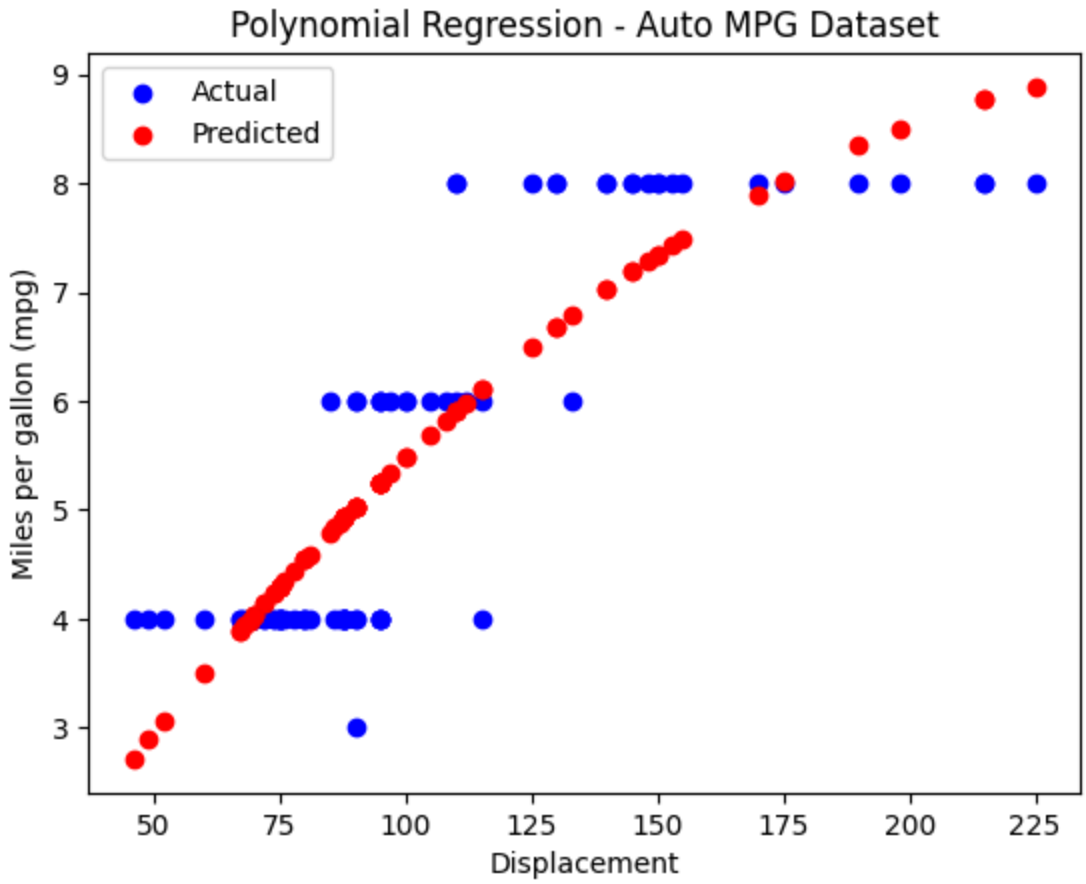
```
#lab_7
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures, StandardScaler
from sklearn.pipeline import make_pipeline
from sklearn.metrics import mean_squared_error, r2_score

def linear_regression_boston():
  url="https://raw.githubusercontent.com/selva86/datasets/master/BostonHousing
  df = pd.read_csv(url)
  X = df[["rm"]].values
  y = df["medv"].values
  X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,rand
  model = LinearRegression()
  model.fit(X_train, y_train)
  y_pred = model.predict(X_test)
  plt.scatter(X_test, y_test, color="blue", label="Actual")
  plt.plot(X_test, y_pred, color="red", label="Predicted")
  plt.xlabel("Average number of rooms (RM)")
  plt.ylabel("Median value of homes ($1000)")
  plt.title("Linear Regression - Boston Housing Dataset")
  plt.legend()
  plt.show()
  print("Linear Regression - Boston Housing Dataset")
  print("Mean Squared Error:", mean_squared_error(y_test, y_pred))
  print("R^2 Score:", r2_score(y_test, y_pred))
```

```
def polynomial_regression_auto_mpg():
  url = "https://archive.ics.uci.edu/ml/machine-learning-databases/auto-mpg/au
  column_names = ["mpg", "cylinders", "displacement", "horsepower", "weight","
  data = pd.read_csv(url, sep=r'\s+', names=column_names, na_values="?")
  data = data.dropna()
  X = data["displacement"].values.reshape(-1, 1)
  y = data["mpg"].values
  X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,rand
  poly_model = make_pipeline(PolynomialFeatures(degree=2), StandardScaler(),Li
  poly_model.fit(X_train, y_train)
  y_pred = poly_model.predict(X_test)
  plt.scatter(X_test, y_test, color="blue", label="Actual")
  plt.scatter(X_test, y_pred, color="red", label="Predicted")
  plt.xlabel("Displacement")
  plt.ylabel("Miles per gallon (mpg)")
  plt.title("Polynomial Regression - Auto MPG Dataset")
  plt.legend()
  plt.show()
  print("Polynomial Regression - Auto MPG Dataset")
  print("Mean Squared Error:", mean_squared_error(y_test, y_pred))
  print("R^2 Score:", r2_score(y_test, y_pred))

if __name__ == "__main__":
  print("Demonstrating Linear Regression and Polynomial Regression\n")
  linear_regression_boston()
  polynomial_regression_auto_mpg()
```

```
Demonstrating Linear Regression and Polynomial Regression
```




```
Linear Regression - Boston Housing Dataset
Mean Squared Error: 46.144775347317264
R^2 Score: 0.3707569232254778
```



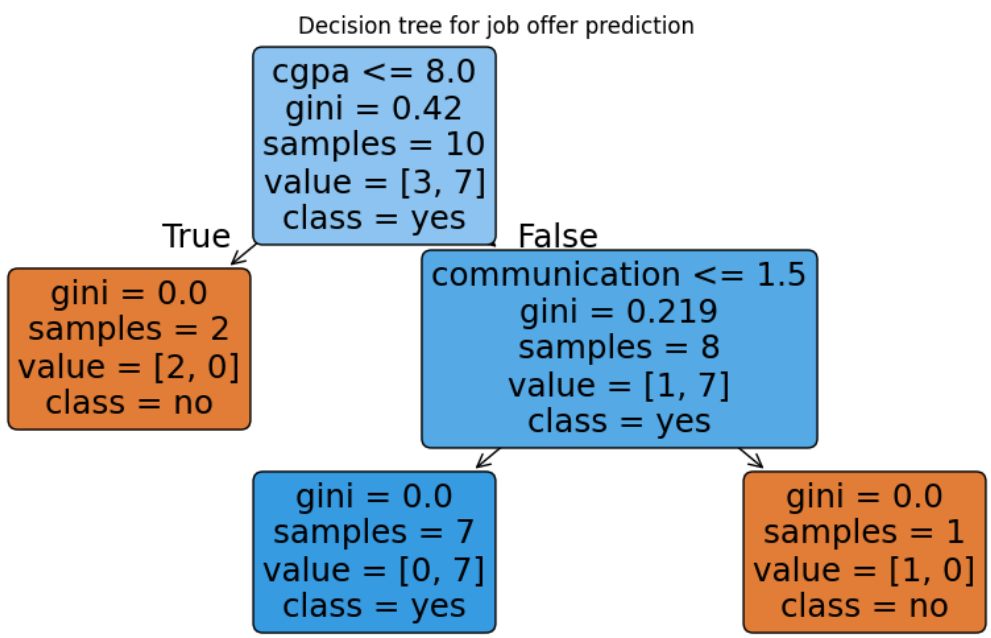

```
Polynomial Regression - Auto MPG Dataset
Mean Squared Error: 0.7431490557205861
R^2 Score: 0.7505650609469626
```

In [ ]:

```
#lab_8

import pandas as pd
import matplotlib.pyplot as plt
from sklearn.datasets import load_breast_cancer
from sklearn.tree import DecisionTreeClassifier,plot_tree
from sklearn.preprocessing import LabelEncoder

cancer=load_breast_cancer()
X_cancer=cancer.data
y_cancer=cancer.target

bc_model=DecisionTreeClassifier(max_depth=4,random_state=42)
bc_model.fit(X_cancer,y_cancer)
print("Model trained successfully")

sample_data=pd.DataFrame({
    'cgpa':[9.2,8.5,9.0,7.5,8.2,9.1,7.8,9.3,8.4,8.6],
    'interactiveness':['yes','no','no','no','yes','yes','yes','yes','no','yes'
    'practical_knowledge':['verygood','good','average','average','good','good'
    'communication':['good','moderate','poor','good','moderate','moderate','po
    'job_offer':['yes','yes','no','no','yes','yes','no','yes','yes','yes']
})
```

```
label_encoders={}
for column in ['interactiveness','practical_knowledge','communication','job_of
  le=LabelEncoder()
  sample_data[column]=le.fit_transform(sample_data[column])
  label_encoders[column]=le

X_sample=sample_data.drop('job_offer',axis=1)
y_sample=sample_data['job_offer']

sample_model=DecisionTreeClassifier(max_depth=4,random_state=42)
sample_model.fit(X_sample,y_sample)

plt.figure(figsize=(12,6))
plot_tree(sample_model,feature_names=X_sample.columns,class_names=label_encode
plt.title("Decision tree for job offer prediction")
plt.show()

test_sample=pd.DataFrame([{
    'cgpa':6.5,
    'interactiveness':'yes',
    'practical_knowledge':'good',
    'communication':'good'
}])

for column in ['interactiveness','practical_knowledge','communication']:
  test_sample[column]=label_encoders[column].transform(test_sample[column])
prediction=sample_model.predict(test_sample)
predicted_label=label_encoders['job_offer'].inverse_transform(prediction)
print("Predicted Job offer for test sample: ",predicted_label[0])
```

```
Model trained successfully
```




```
Predicted Job offer for test sample:  no
```

In [ ]:

```
#lab_9

#LAB9
import numpy as np
from sklearn.datasets import fetch_olivetti_faces
from sklearn.model_selection import train_test_split,cross_val_score
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score,classification_report,confusion_mat
import matplotlib.pyplot as plt

data=fetch_olivetti_faces(shuffle=True,random_state=42)
X=data.data
y=data.target

X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.3,random_state=

gnb=GaussianNB()

gnb.fit(X_train,y_train)

y_pred=gnb.predict(X_test)

accuracy=accuracy_score(y_test,y_pred)
print(f'Accuracy:{accuracy*100:.2f}%')

print("\nClassification ReportL ")
print(classification_report(y_test,y_pred,zero_division=1))

print("\n Confusion Matrix: ")
print(confusion_matrix(y_test,y_pred))

fig,axes=plt.subplots(3,5,figsize=(12,8))
for ax,image,label,prediction in zip(axes.ravel(),X_test,y_test,y_pred):
  ax.imshow(image.reshape(64,64),cmap=plt.cm.gray)
  ax.set_title(f"True:{label} Pred:{prediction}")
  ax.axis('off')

plt.tight_layout()
plt.show()
```

Accuracy:89.17%

Classification ReportL

	precision	recall	f1-score	support
0	1.00	0.67	0.80	3
1	1.00	0.67	0.80	3
2	0.67	0.67	0.67	3
3	0.43	1.00	0.60	3
4	1.00	1.00	1.00	3
5	1.00	1.00	1.00	3
6	1.00	0.67	0.80	3
7	0.75	1.00	0.86	3
8	1.00	1.00	1.00	3
9	1.00	0.67	0.80	3
10	1.00	1.00	1.00	3
11	1.00	1.00	1.00	3
12	1.00	0.33	0.50	3
13	1.00	1.00	1.00	3
14	1.00	1.00	1.00	3
15	0.75	1.00	0.86	3
16	0.60	1.00	0.75	3
17	1.00	1.00	1.00	3
18	1.00	1.00	1.00	3
19	1.00	1.00	1.00	3
20	0.75	1.00	0.86	3
21	1.00	1.00	1.00	3
22	1.00	1.00	1.00	3
23	1.00	1.00	1.00	3
24	1.00	0.67	0.80	3
25	1.00	0.67	0.80	3
26	1.00	0.67	0.80	3
27	1.00	1.00	1.00	3
28	1.00	1.00	1.00	3
29	1.00	0.67	0.80	3
30	0.75	1.00	0.86	3
31	1.00	0.67	0.80	3
32	1.00	1.00	1.00	3
33	1.00	0.67	0.80	3
34	1.00	1.00	1.00	3
35	1.00	1.00	1.00	3
36	1.00	1.00	1.00	3
37	1.00	1.00	1.00	3
38	1.00	1.00	1.00	3
39	0.60	1.00	0.75	3
accuracy			0.89	120
macro avg	0.93	0.89	0.89	120
weighted avg	0.93	0.89	0.89	120

Confusion Matrix:

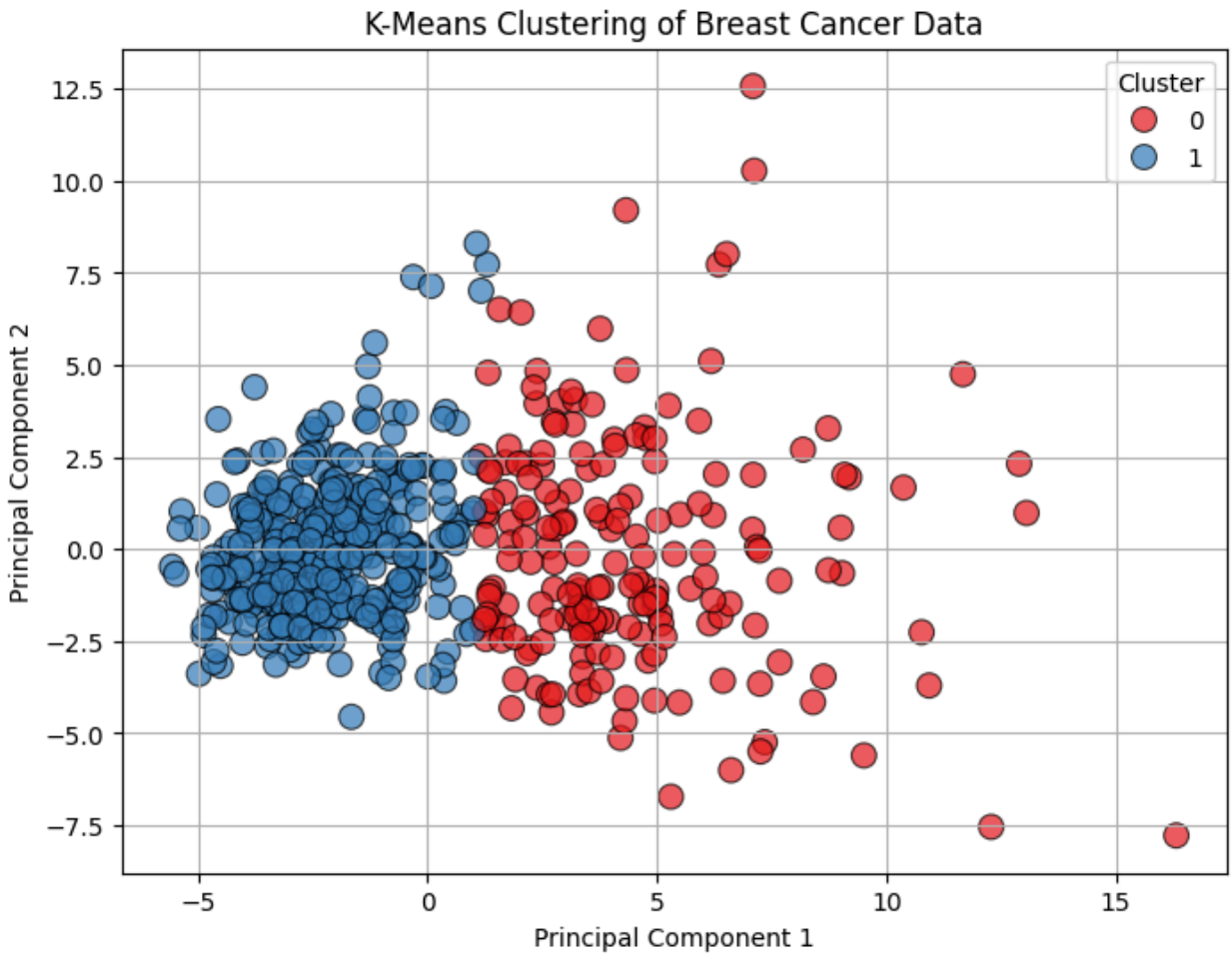
```
[[2 0 0 ... 0 0 0]
 [0 2 0 ... 0 0 0]
```

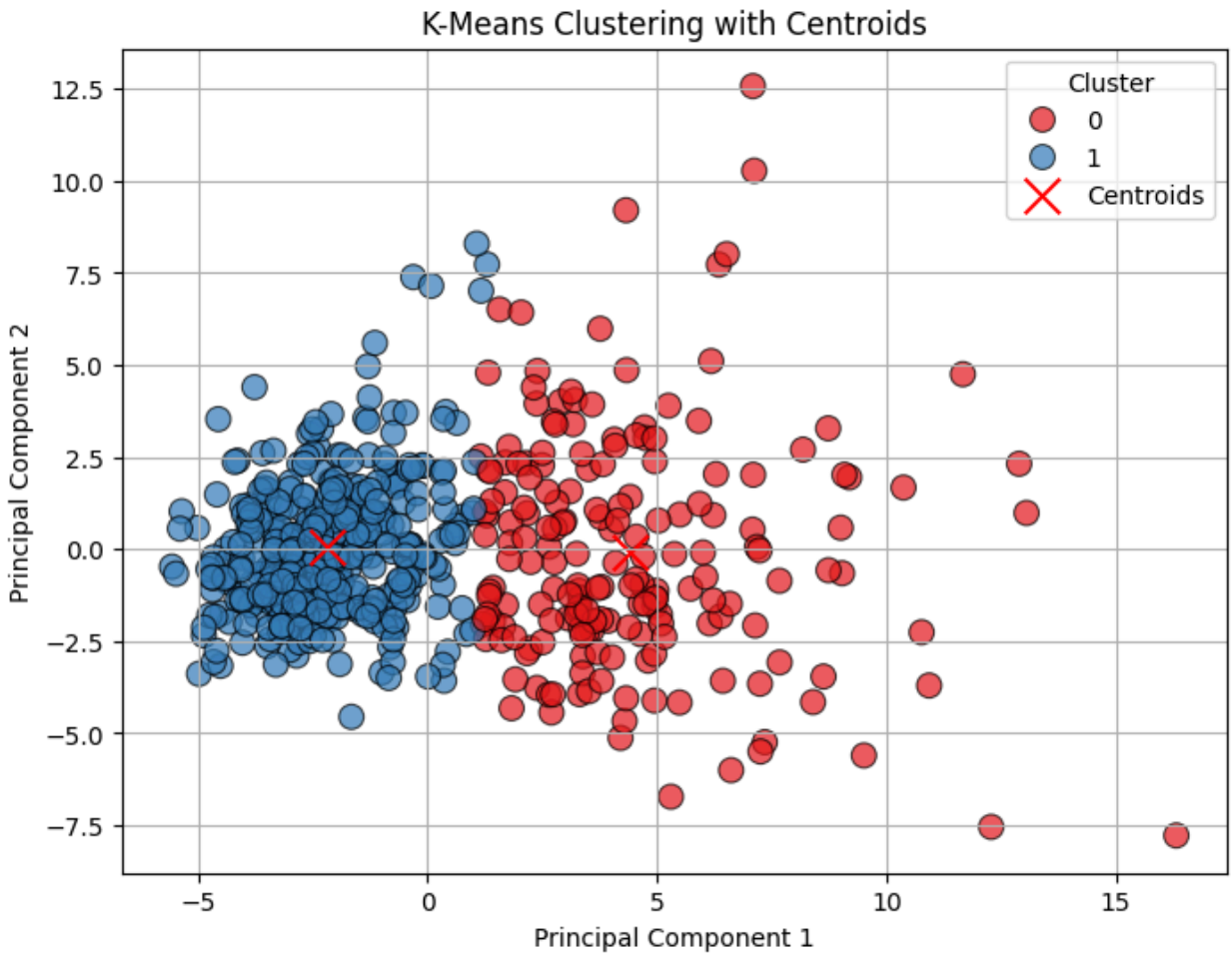
```
[0 0 2 ... 0 0 1]
...
[0 0 0 ... 3 0 0]
[0 0 0 ... 0 3 0]
[0 0 0 ... 0 0 3]]
```

In [ ]:

```python
#lab_10

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.datasets import load_breast_cancer
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
from sklearn.metrics import confusion_matrix, classification_report

data=load_breast_cancer()
X=data.data
y=data.target

scaler=StandardScaler()
X_scaled=scaler.fit_transform(X)

kmeans=KMeans(n_clusters=2,random_state=42)

y_kmeans=kmeans.fit_predict(X_scaled)
```

```
pca=PCA(n_components=2)
X_pca=pca.fit_transform(X_scaled)

df=pd.DataFrame(X_pca,columns=['PC1','PC2'])
df['Cluster']=y_kmeans
df['True Label']=y

plt.figure(figsize=(8,6))
sns.scatterplot(data=df,x='PC1',y='PC2',hue='Cluster',palette='Set1',s=100,edg
plt.title('K-Means Clustering of Breast Cancer Data')
plt.xlabel('Principal Component 1')
plt.ylabel('Principal Component 2')
plt.legend(title='Cluster')
plt.grid(True)
plt.show()

plt.figure(figsize=(8,6))
sns.scatterplot(data=df,x='PC1',y='PC2',hue='Cluster',palette='Set1',s=100,edg
centers=pca.transform(kmeans.cluster_centers_)
plt.scatter(centers[:,0],centers[:,1],c='red',marker='x',s=200,label='Centroid
plt.title('K-Means Clustering with Centroids')
plt.xlabel('Principal Component 1')
plt.ylabel('Principal Component 2')
plt.legend(title='Cluster')
plt.grid(True)
plt.show()
```


K-Means Clustering of Breast Cancer Data
Cluster
0
1
Principal Component 2
Principal Component 1
12.5
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
−5
0
5
10
15


K-Means Clustering with Centroids
Cluster
0
1
Centroids
Principal Component 2
Principal Component 1
12.5
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
−5
0
5
10
15